

UNIT 4: Centroid and Moment of Inertia

Syllabus topics: Centroid of plane areas & composite bodies; moment of inertia of plane area; parallel axis theorem & perpendicular axis theorem; radius of gyration; principal moment of inertia.

Question 1

3 Determine the coordinates of the centroid of the shaded area.

5	CO4	K2, K3, K5
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Course Outcomes	CO1	Understand the basic laws concepts of mechanical systems.
	CO2	Determine resultants and apply conditions of static equilibrium to plane force systems.
	CO3	Analyze beam for shear force and bending moment along the span and analyze trusses for axial forces
	CO4	Evaluate the structural properties centroid and moment of inertia
	CO5	Stress analysis for one- and two-dimensional stress systems.

MID TERM-II | NME 101 | Q3 Centroid of trapezoid with circular cut-out

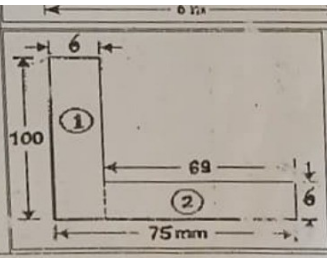
Question 2

Figure 2.

CO1	Understand the basic laws concepts of mechanical systems.	K1, K2
CO2	Determine resultants and apply conditions of static equilibrium to plane force systems.	K1, K3

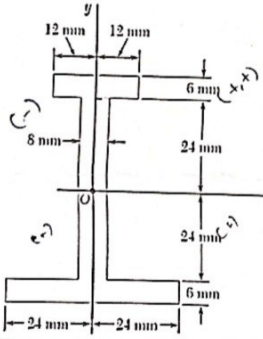
MID TERM-II | NME 102 | 2025-26 | Q2(b) Centroid (C.G.) of uniform plane lamina

Question 3

3		Calculate Centroid of following diagram (All dimensions are in mm).	5	CO4	K3, K4
	CO1	Understand the basic laws concepts of mechanical systems.			
	CO2	Determine resultants and apply conditions of static equilibrium to plane force systems.			
	CO3	Analyze beam for shear force and bending moment along the span and analyze trusses for axial			

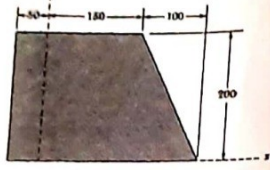
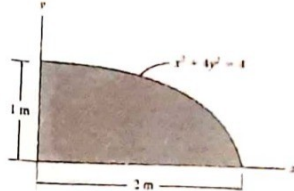
2nd MID SEM | NME-101 | 2024-25 | Q3 Centroid of L-shaped section

Question 4

4a	State and prove the parallel-axis theorem	5	CO4	K1, K2
4b	Determine the moment of inertia and the radius of gyration of the I-section area with respect to the x-axis.	5	CO4	K2, K3, K4
				
5a	State the Hooke's law and draw the typical stress-strain curve for a ductile and brittle material.	3	CO5	K1, K2

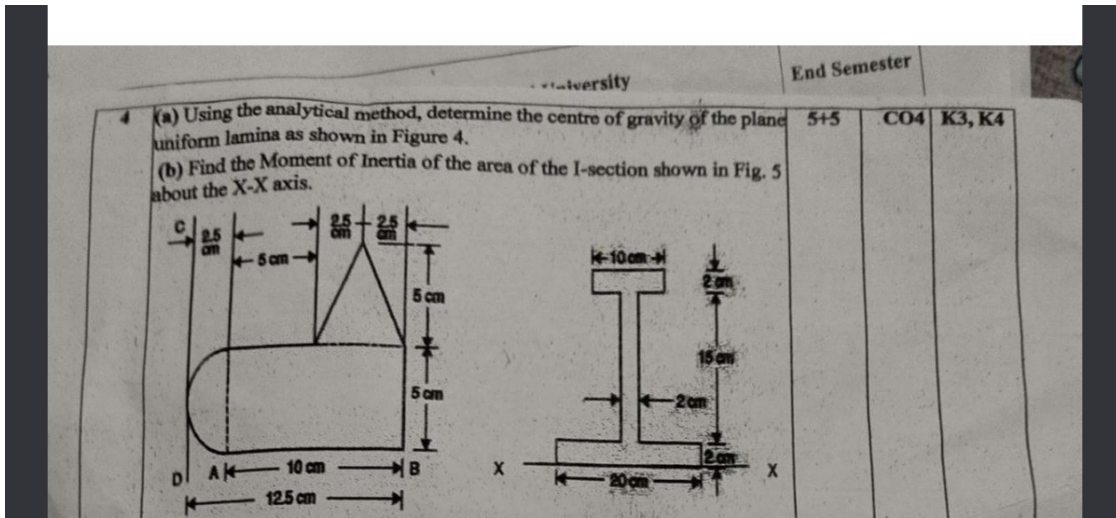
NME 101 | 2025-26 | Q4(a) Parallel-axis theorem (state & prove), Q4(b) Moment of inertia & radius of gyration of I-section

Question 5

4(a)	Determine the coordinates of the centroid of the shaded area with respect to given X-Y axis. (Refer Figure 4)	5	CO4	K4, K5
4(b)	Determine the moment of Inertia of the shaded area as shown in figure-5 with respect to both x and y axis.	5	CO4	K4, K5
	 Dimensions in millimeters Figure 4			
	 Figure 5			

END SEM | NME-102 | 2022-23 | Q4(a) Centroid of shaded area, Q4(b) Moment of inertia of shaded area about x & y axis

Question 6



End Semester | Q4(a) Centre of gravity of plane lamina, Q4(b) Moment of inertia of I-section about X-X axis